

# Ultimatum Game Teacher Notes

TOK Cognitive Science Activity Bank • Human Sciences • 25-35 min

|            |                |      |                |
|------------|----------------|------|----------------|
| Category   | Human Sciences | Time | 25-35 min      |
| Difficulty | Medium         | ID   | ultimatum-game |

Big idea: Humans do not always maximise narrow utility.

## Overview

Students test whether human decisions always match narrow economic rationality.

## Student Prompt

Inspect Ultimatum Game Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

## Concrete Stimulus Packet

|                                      |                                                                                                                                                                                                                          |                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Ultimatum Game Stimulus A</b>     | Student-facing stimulus 1: Students split ten tokens and record accept/reject decisions. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.           | Students inspect this exact card first and commit to a claim before hearing anyone else.                     |
| <b>Ultimatum Game Stimulus B</b>     | Student-facing stimulus 2: Offer cards showing fair, unfair, strategic, and emotional proposals. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.   | Use this for comparison, a second round, or a partner challenge.                                             |
| <b>Ultimatum Game Reveal / Twist</b> | Student-facing stimulus 3: A graph comparing rational-choice prediction with actual class choices. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result. | Teacher reveal: connect the changed response to the big idea — Humans do not always maximise narrow utility. |

## Actual Lesson Questions and Key

| # | Question for students                                                           | Teacher key / cue                                                                                             |
|---|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 | After inspecting Ultimatum Game Stimulus A, what is your first knowledge claim? | Teacher cue: look for a clear claim, not just a description.                                                  |
| 2 | Which exact detail from Ultimatum Game Stimulus A did you use as evidence?      | Teacher cue: students should name visible words, data, source features, method features, or contextual clues. |
| 3 | What did you infer beyond the evidence?                                         | Teacher cue: this is where assumptions, framing, models, or perspective usually appear.                       |
| 4 | How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?                        | Teacher cue: confidence is data for the debrief, not a grade.                                                 |
| 5 | What missing information would most improve the knowledge claim?                | Teacher cue: push for method, source, context, comparison, or definitions.                                    |
| 6 | Can human behaviour be predicted scientifically?                                | Teacher cue: students should move from the classroom example to a general TOK issue.                          |

## Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how humans do not always maximise narrow utility.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

## Setup Checklist

- Prepare a starter stimulus using: Students split ten tokens and record accept/reject decisions.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

## Ready-to-Use Examples

- Students split ten tokens and record accept/reject decisions.
- Offer cards showing fair, unfair, strategic, and emotional proposals.
- A graph comparing rational-choice prediction with actual class choices.

## Suggested Timing

| Stage   | Teacher move                                                | Watch for                                                                   |
|---------|-------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1       | Pair students. Player A divides 10 tokens between A and B.  | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 2       | Player B accepts or rejects. If rejected, both get nothing. | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 3       | Collect anonymous results.                                  | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 4       | Discuss why people reject unfair offers.                    | Assumptions, confidence shifts, evidence claims, and language choices.      |
| Debrief | Move from the activity result to a TOK knowledge question.  | Students distinguishing method, evidence, interpretation, and overclaiming. |

## Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

## Common Misconceptions

- Students may treat class behaviour as universal human nature.
- Students may ignore ethical limits or context effects in human-subject inquiry.

## Assessment Look-Fors

- Connects behaviour to method, context, incentives, and ethical constraints.
- Avoids overgeneralising from one small sample.

## **Differentiation and Adaptations**

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

## **Extension Tasks**

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

## **Related Activities**

- Survey Wording Experiment
- Conformity without Humiliation
- Replication Debate